

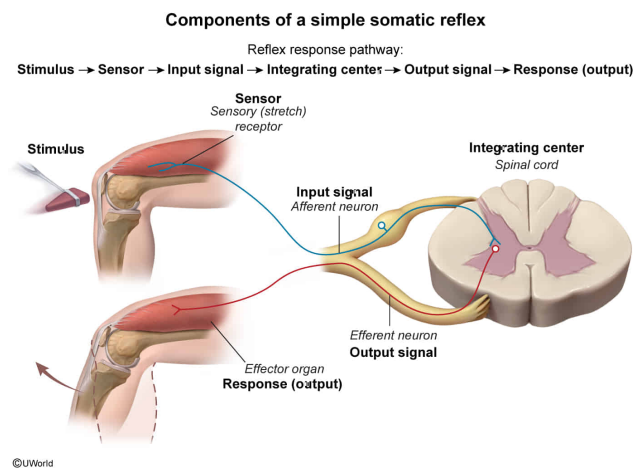
Somatic neural reflexes promote involuntary responses of skeletal muscles to specific stimuli. Given this, a somatic neural reflex requires:

- ✓ ☒ A. an afferent neuron. [23%]
- ☐ B. an autonomic neuron. [21%]
- ☐ C. both an afferent and an autonomic neuron. [49%]
- ☐ D. neither an afferent nor an autonomic neuron. [4%]

Correct

22% answered correctly

Explanation:



Reflexes are involuntary responses to specific stimuli. Some reflexes involve **neurons** eliciting a response in a target tissue, such as when a **finger is quickly withdrawn** from a hot surface. Other reflexes do not involve neurons, such as when an **endocrine tissue** adjusts hormone production based on the level of another hormone in the blood. **Neuroendocrine reflexes** involve both neurons and endocrine cells.

Neural reflexes (ie, the reflexes in which neurons play a part) can be either **somatic** (involving a **skeletal muscle**) or **autonomic** (involving tissues other than skeletal muscle). Both somatic and autonomic reflexes include an **afferent neuron** carrying sensory information toward the central nervous system (CNS) and an **efferent neuron** that activates a response in a target tissue. In some somatic (and all autonomic) reflexes, an additional neuron is included within the CNS between the afferent and efferent neurons. Such neurons, called **interneurons**, can modulate the reflex and send information to and from the brain.

The question states that neural reflexes involve involuntary responses to stimuli and asks about the requirements for a neural reflex that activates a skeletal

muscle to occur. The **simplest** (ie, **monosynaptic**) **somatic reflexes only require** an **afferent neuron** delivering the sensory information to the spinal cord **and an efferent motor neuron** activating the muscle response. Polysynaptic reflexes incorporate at least one more neuron in the reflex.

(Choices B, C, and D) Some neural reflexes activate effector responses by tissues other than skeletal muscle (eg, glands, smooth muscle), and these reflexes engage autonomic effector neurons. However, somatic reflexes do *not* include autonomic neurons.

Educational objective:

Reflexes are involuntary responses to specific stimuli and can be either neural, endocrine, or neuroendocrine. Neural reflexes include both somatic and autonomic reflexes. Somatic reflexes can be monosynaptic or polysynaptic whereas autonomic reflexes are always polysynaptic.

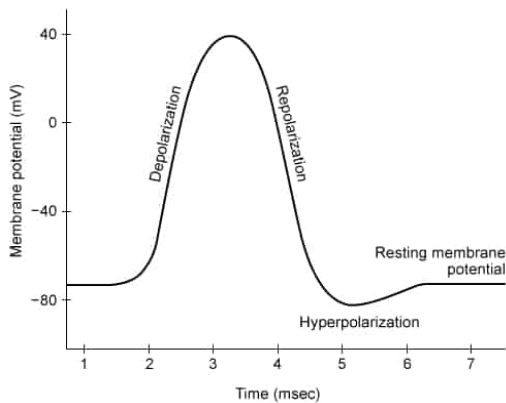
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Subject:
Biology

QID:404165

System:
Endocrine and Nervous Systems

Below is a diagram of an action potential. Which statement best explains why another depolarization cannot occur during the repolarization period?

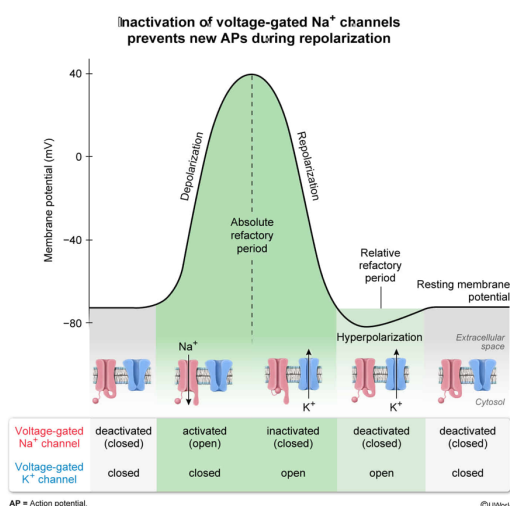


- ☐ A. The membrane potential must fully return to a resting state before another depolarization can occur. [25%]
- ☐ B. Na^+ channels are prevented from re-opening during repolarization by the opening of K^+ channels. [50%]
- ☐ C. Both K^+ channels and Na^+ channels must be inactivated during repolarization. [1%]
- ☒ D. During repolarization, Na^+ channel inactivation gates are closed. [21%]

Correct

20% answered correctly

Explanation:



An **action potential** (AP) is a change in the membrane potential of a **neuron** and results in the transmission of an electrical signal down its **axon**. APs take place in three stages:

- 1. Depolarization:** The **resting membrane potential** is maintained at -70 mV. Voltage-gated Na^+ **channels** are deactivated (closed) at rest, but when a stimulus reaches a **threshold** level (-55 mV), the activation gates open and Na^+ ions enter the cell.
- 2. Repolarization:** As the membrane potential reaches +30 mV (overshoot), the inactivation gates of Na^+ channels close, preventing re-entry of Na^+ ions, and K^+ channels are opened. The Na^+ channel inactivation gates remain closed while the cell repolarizes.

3. **Hyperpolarization:** Membrane potential becomes slightly more negative than resting potential (undershoot), until K^+ concentration reaches equilibrium. The resting membrane potential is fully restored when K^+ channels close. Na^+ channels return to a deactivated (closed) state during this time.

The question asks why another AP cannot occur during repolarization. Repolarization begins when K^+ channels open, causing K^+ ions to rapidly leave the cell. Simultaneously, the inactivation gates of the Na^+ channels close, preventing the re-entry of Na^+ ions. The **absolute refractory period** is the period during which no new APs can be generated, regardless of the strength of stimulus received.

The closing of Na^+ channel inactivation gates prevents a new AP from occurring during repolarization. To begin a new AP, the Na^+ channels must return to a deactivated (closed) state.

(Choice A) During hyperpolarization, a new AP *can* be generated, even though the membrane is *below* resting potential, because inactivation gates have been opened, but a stronger than normal stimulus would be needed. This period is called the relative refractory period.

(Choices B and C) Although a new AP cannot occur during depolarization (when Na^+ channel activation gates are open and K^+ channels are closed) or during repolarization (when Na^+ channel inactivation gates are closed), it *can* occur while K^+ channels are still open (the relative refractory period) if the stimulus is strong enough.

Educational objective:

During an action potential (AP), voltage-gated Na^+ and K^+ channels are opened and closed in a sequential manner to propagate an electrical signal down the axon. During the absolute refractory period, a new AP cannot be generated. The Na^+ channel inactivation gates must open and the Na^+ channels must return to a deactivated (closed) state before the next AP can occur.

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